

Problem 1.

(S.o.l.)
 1° In Regularized MDP, Bellman optimal Equation:

$$V_{\pi}^*(s) = \max_{\pi \in \Pi} Q_{\pi}^*(s, a) = \max_{\pi \in \Pi} R_s^a + \gamma P_s^{\pi} V_{\pi}^*(s) \quad \text{--- ①}$$

$$Q_{\pi}^*(s, a) = R_s^a + \gamma \bar{E}_{s' \sim P(\cdot|s, a)} [V_{\pi}^*(s')] \quad \text{--- ②}$$

Substitute ① into ②: $Q_{\pi}^*(s, a) = R_s^a + \gamma \bar{E}_{s' \sim P(\cdot|s, a)} \left[\max_{\pi' \in \Pi} \left(\pi(1|s') \cdot Q_{\pi'}^*(s, a) + \alpha H(\pi(1|s')) \right) \right] = Q_{\pi_{s, t}}^*(s, a)$

⇒ we get $Q_{s, t}^*$, can use this to evaluate policy
 $\tilde{\pi}_{t+1}(1|s) = \arg \max_{\pi \in \Pi} [\pi(1|s) \cdot Q_{s, t}^{\tilde{\pi}}(s, a) + \alpha H(\pi(1|s))]$, in this problem, $\alpha=1$, $H(\pi(1|s)) = \Omega(\pi(1|s))$
 $= \bar{\sum}_{a \in A} \pi(a|s) \cdot \log \pi(a|s)$

We can convert the problem into following Optimization Problem:
 $\max_{\pi(1|s) \in A} \bar{\sum}_{a \in A} [\pi(a|s) \cdot Q_{s, t}^{\pi}(s, a) - \pi(a|s) \cdot \log \pi(a|s)]$, subject to $\bar{\sum}_{a \in A} \pi(a|s) - 1 = 0$

2° We leverage Lagrange multiplier technique, set μ as Lagrange multiplier.

⇒ Then we can construct the Lagrangian as:

$$L(\pi) = \bar{\sum}_{a \in A} (\pi(a|s) \cdot Q_{s, t}^{\pi}(s, a) - \pi(a|s) \cdot \log \pi(a|s)) - \mu (\bar{\sum}_{a \in A} \pi(a|s) - 1)$$
, finding the optimal solution.

we let $\frac{\partial L(\pi)}{\partial \pi(a|s)} = 0$, for every $a \in A$.

$$3^{\circ} \frac{\partial L(\pi)}{\partial \pi(a|s)} = Q_{s, t}^{\pi}(s, a) - \log \pi(a|s) - 1 - \mu \stackrel{\text{let } 0}{=} 0.$$

$$\Rightarrow Q_{s, t}^{\pi}(s, a) - 1 - \mu = \log \pi(a|s)$$

$$\Rightarrow \pi(a|s) = e^{Q_{s, t}^{\pi}(s, a) - 1 - \mu}$$
, since $e^{-1-\mu}$ is constant, we let it be C_s .

$$\Rightarrow \pi(a|s) = C_s \cdot e^{Q_{s, t}^{\pi}(s, a)}$$
 Remind the constraint: $\bar{\sum}_{a \in A} \pi(a|s) = 1$.

$$\Rightarrow \bar{\sum}_{a \in A} \pi(a|s) = \bar{\sum}_{a \in A} C_s \cdot e^{Q_{s, t}^{\pi}(s, a)} = 1, \quad C_s = \frac{1}{\bar{\sum}_{a \in A} e^{Q_{s, t}^{\pi}(s, a)}}$$

$$\Rightarrow \pi(1|s) = \frac{e^{Q_{s, t}^{\pi}(s, 1)}}{\bar{\sum}_{a \in A} e^{Q_{s, t}^{\pi}(s, a)}}$$